

# 电动力学作业

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## 第五章

4.(1) 因为真空中无电荷和电流分布, 有:

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} = \frac{\partial \mathbf{D}}{\partial t}$$

又因为  $\mathbf{D} = \epsilon_0 \mathbf{E}$ ,  $\mathbf{B} = \mu_0 \mathbf{H}$ , 故:

$$\frac{1}{\mu_0} \nabla \times \mathbf{B} = \epsilon_0 \frac{\partial \mathbf{D}}{\partial t}$$

代入电场的定义  $\mathbf{E} = -\nabla\varphi - \frac{\partial \mathbf{A}}{\partial t}$  和磁矢势的定义  $\mathbf{B} = \nabla \times \mathbf{A}$  得: (注意  $\mu_0 \epsilon_0 = \frac{1}{c^2}$ )

$$\frac{1}{\mu_0} \nabla \times (\nabla \times \mathbf{A}) = \epsilon_0 \frac{\partial}{\partial t} \left( -\nabla\varphi - \frac{\partial \mathbf{A}}{\partial t} \right)$$

在 Lorentz 规范  $\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$  下:

$$\frac{1}{\mu_0} \nabla \times (\nabla \times \mathbf{A}) = \frac{1}{\mu_0} \nabla(\nabla \cdot \mathbf{A}) - \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2}$$

又用公式  $\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A})$  化简得:

$$\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0$$

这便是  $\mathbf{A}$  满足的波动方程。

注意到:

$$\begin{aligned} \nabla^2(e^{i\mathbf{k} \cdot \mathbf{x}}) &= (\partial_x^2 + \partial_y^2 + \partial_z^2) \cdot \left( e^{ik_x x + ik_y y + ik_z z} \right) \\ &= [(ik_x)^2 + (ik_y)^2 + (ik_z)^2] e^{i\mathbf{k} \cdot \mathbf{x}} \\ &= -k^2 e^{i\mathbf{k} \cdot \mathbf{x}} \end{aligned}$$

同理有:

$$\nabla^2(e^{-i\mathbf{k} \cdot \mathbf{x}}) = -k^2 e^{-i\mathbf{k} \cdot \mathbf{x}}$$

另一方面,  $\mathbf{A}$  对时间的二阶导数为:

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} = \left( \frac{d^2 \mathbf{a}_k}{dt^2} e^{i\mathbf{k} \cdot \mathbf{x}} + \frac{d^2 \mathbf{a}_k^*}{dt^2} e^{-i\mathbf{k} \cdot \mathbf{x}} \right)$$

代入  $\mathbf{A}$  的波动方程得：

$$\sum_k \left( \mathbf{a}_k e^{i\mathbf{k}\cdot\mathbf{x}} + \mathbf{a}_k^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right) + \frac{1}{c^2} \sum_k \left( \frac{d^2 \mathbf{a}_k}{dt^2} e^{i\mathbf{k}\cdot\mathbf{x}} + \frac{d^2 \mathbf{a}_k^*}{dt^2} e^{-i\mathbf{k}\cdot\mathbf{x}} \right) = 0$$

对第  $k$  项的系数则有：

$$\frac{d^2 \mathbf{a}_k}{dt^2} + k^2 c^2 \mathbf{a}_k = 0, \quad \frac{d^2 \mathbf{a}_k^*}{dt^2} + k^2 c^2 \mathbf{a}_k^* = 0$$

4.(2) 如果选取 Coulomb 规范  $\nabla \cdot \mathbf{A} = 0$  和  $\varphi = 0$ ，代入

$$\mathbf{A} = \sum_k \left( \mathbf{a}_k e^{i\mathbf{k}\cdot\mathbf{x}} + \mathbf{a}_k^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right)$$

得：

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \partial_x A_x + \partial_y A_y + \partial_z A_z \\ &= \sum_{j=x,y,z} \left( ik_j \sum_k a_{kj} e^{i\mathbf{k}\cdot\mathbf{x}} - ik_j \sum_k a_{kj}^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right) \\ &= i \sum_k \left( \mathbf{k} \cdot \mathbf{a}_k e^{i\mathbf{k}\cdot\mathbf{x}} - \mathbf{k} \cdot \mathbf{a}_k^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right) \\ &= 0 \end{aligned}$$

为使得上式恒成立，则有：

$$\mathbf{k} \cdot \mathbf{a}_k = 0, \quad \mathbf{k} \cdot \mathbf{a}_k^* = 0$$

4.(3) 在 Coulomb 规范  $\nabla \cdot \mathbf{A} = 0$  和  $\varphi = 0$  下，有：

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$$

则有：

$$\mathbf{B} = \nabla \times \mathbf{A} = i\mathbf{k} \times \mathbf{A} = \sum_k \left( i\mathbf{k} \times \mathbf{a}_k e^{i\mathbf{k}\cdot\mathbf{x}} + i\mathbf{k} \times \mathbf{a}_k^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right)$$

对于电场：

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = -\sum_k \left( \frac{d\mathbf{a}_k}{dt} e^{i\mathbf{k}\cdot\mathbf{x}} + \frac{d\mathbf{a}_k^*}{dt} e^{-i\mathbf{k}\cdot\mathbf{x}} \right)$$

可以用电磁波的横波性得到：（ $\hat{k}$  为波矢  $\mathbf{k}$  方向的单位矢量）

$$\begin{aligned} \mathbf{E} &= c\mathbf{B} \times \hat{k} \\ &= ic(\mathbf{k} \times \mathbf{A}) \times \hat{k} \\ &= ick(\hat{k} \times \mathbf{A}) \times \hat{k} \\ &= ick \left[ (\hat{k} \cdot \hat{k})\mathbf{A} - (\hat{k} \cdot \mathbf{A})\hat{k} \right] \\ &= ick\mathbf{A} \\ &= \sum_k ick \left( \mathbf{a}_k e^{i\mathbf{k}\cdot\mathbf{x}} + \mathbf{a}_k^* e^{-i\mathbf{k}\cdot\mathbf{x}} \right) \end{aligned}$$

注意这里的  $\mathbf{A}$  与  $\mathbf{E}$  在同一直线上，并与  $\hat{k}$  垂直。

5.(1) 由于  $\varphi = -\nabla \cdot \mathbf{Z}$ , 在 Lorentz 规范  $\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$  下:

$$\nabla \cdot \mathbf{A} - \frac{1}{c^2} \frac{\partial \varphi}{\partial t} (\nabla \cdot \mathbf{Z}) = \nabla \cdot \left( \mathbf{A} - \frac{1}{c^2} \frac{\partial \mathbf{Z}}{\partial t} \right) = 0$$

则  $\mathbf{A} - \frac{1}{c^2} \frac{\partial \mathbf{Z}}{\partial t}$  为一个无旋场, 可取为零, 即:

$$\mathbf{A} = \frac{1}{c^2} \frac{\partial \mathbf{Z}}{\partial t}$$

5.(2) 令  $\rho = -\nabla \cdot \mathbf{P}$ , 则在 Lorentz 规范下:

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\epsilon_0}$$

则有:

$$\nabla^2 (-\nabla \cdot \mathbf{Z}) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (-\nabla \cdot \mathbf{Z}) = -\frac{1}{\epsilon_0} (-\nabla \cdot \mathbf{P})$$

即:

$$\nabla \cdot \left( \nabla^2 \mathbf{Z} - \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} + \frac{\mathbf{P}}{\epsilon_0} \right) = 0$$

则  $\nabla^2 \mathbf{Z} - \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} + \frac{\mathbf{P}}{\epsilon_0}$  为无旋场, 可取为 0, 故:

$$\nabla^2 \mathbf{Z} - \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} = -\frac{\mathbf{P}}{\epsilon_0}$$

这个方程与  $\mathbf{A}$  满足的 d'Alembert 方程有相同的形式,  $\mathbf{Z}$  满足的推迟解为:

$$\mathbf{Z}(\mathbf{x}, t) = \frac{1}{4\pi\epsilon_0} \iiint \frac{1}{r} \mathbf{P} \left( \mathbf{x}', t - \frac{r}{c} \right) dV'$$

5.(3) 由  $\varphi = -\nabla \cdot \mathbf{Z}$  和  $\mathbf{A} = \frac{1}{c^2} \frac{\partial \mathbf{Z}}{\partial t}$  得:

$$\begin{aligned} \mathbf{E} &= -\nabla \varphi - \frac{\partial \mathbf{A}}{\partial t} \\ &= \nabla (\nabla \cdot \mathbf{Z}) - \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} \\ &= \nabla^2 \mathbf{Z} + \nabla \times (\nabla \times \mathbf{Z}) - \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} \end{aligned}$$

又因为:

$$\nabla^2 \mathbf{Z} = \frac{1}{c^2} \frac{\partial^2 \mathbf{Z}}{\partial t^2} - \frac{\mathbf{P}}{\epsilon_0}$$

则有:

$$\mathbf{E} = \nabla \times (\nabla \times \mathbf{Z}) - \frac{\mathbf{P}}{\epsilon_0}$$

磁场为:

$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A} \\ &= \nabla \times \left( \frac{1}{c^2} \frac{\partial \mathbf{Z}}{\partial t} \right) \\ &= \frac{1}{c^2} \frac{\partial}{\partial t} (\nabla \times \mathbf{Z}) \end{aligned}$$

9. 由题意,  $\mathbf{A}$  和  $\varphi$  的推迟势为:

$$\begin{aligned}\mathbf{A}(\mathbf{x}, t) &= \frac{\mu_0}{4\pi} \iiint \frac{1}{r} \mathbf{J}(\mathbf{x}', t') dV' \\ \varphi(\mathbf{x}, t) &= \frac{1}{4\pi\epsilon_0} \iiint \frac{1}{r} \rho(\mathbf{x}', t') dV'\end{aligned}$$

其中  $t' = t - \frac{r}{c}$ 。

首先从  $\nabla' \cdot \mathbf{J}$  入手:

$$\nabla' \cdot \mathbf{J}(\mathbf{x}', t') = \nabla' \cdot \mathbf{J}(\mathbf{x}') + \frac{\partial \mathbf{J}(\mathbf{x}', t')}{\partial t'} \cdot \nabla' t'$$

由  $t' = t - \frac{r}{c}$  得  $\nabla t' = -\frac{\nabla r}{c}$ , 以及算符代换关系  $\nabla' \rightarrow -\nabla$ , 有:

$$\begin{aligned}\nabla' \cdot \mathbf{J}(\mathbf{x}', t') &= \nabla' \cdot \mathbf{J}(\mathbf{x}') - \frac{\partial \mathbf{J}(\mathbf{x}', t')}{\partial t'} \cdot \nabla t' \\ &= \nabla' \cdot \mathbf{J}(\mathbf{x}') + \frac{\partial \mathbf{J}(\mathbf{x}', t')}{\partial t'} \cdot \frac{\nabla r}{c} \\ &= \nabla' \cdot \mathbf{J}(\mathbf{x}') - \nabla \cdot \mathbf{J}(\mathbf{x}', t')\end{aligned}$$

于是有:

$$\begin{aligned}\nabla' \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right] &= \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}', t')] + \left( \nabla' \frac{1}{r} \right) \cdot \mathbf{J}(\mathbf{x}', t') \\ &= \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}', t')] - \left( \nabla \frac{1}{r} \right) \cdot \mathbf{J}(\mathbf{x}', t') \\ &= \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] - \frac{1}{r} [\nabla \cdot \mathbf{J}(\mathbf{x}', t')] - \left( \nabla \frac{1}{r} \right) \cdot \mathbf{J}(\mathbf{x}', t')\end{aligned}$$

即:

$$\frac{1}{r} [\nabla \cdot \mathbf{J}(\mathbf{x}', t')] + \left( \nabla \frac{1}{r} \right) \cdot \mathbf{J}(\mathbf{x}', t') = \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] - \nabla' \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right]$$

进一步得到:

$$\begin{aligned}\nabla \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right] &= \frac{1}{r} [\nabla \cdot \mathbf{J}(\mathbf{x}', t')] + \left( \nabla \frac{1}{r} \right) \cdot \mathbf{J}(\mathbf{x}', t') \\ &= \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] - \nabla' \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right]\end{aligned}$$

则:

$$\begin{aligned}\nabla \cdot \mathbf{A}(\mathbf{x}, t) &= \frac{\mu_0}{4\pi} \iiint \nabla \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right] dV' \\ &= \frac{\mu_0}{4\pi} \iiint \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] dV' - \frac{\mu_0}{4\pi} \iiint \nabla' \cdot \left[ \frac{1}{r} \mathbf{J}(\mathbf{x}', t') \right] dV' \\ &= \frac{\mu_0}{4\pi} \iiint \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] dV' - \frac{\mu_0}{4\pi} \oint \frac{1}{r} \mathbf{J}(\mathbf{x}', t') d\mathbf{S}'\end{aligned}$$

最后一步中运用了 Gauss 散度定理将第二项的体积分化为面积分, 而积分面总可以取为大于电流分布区域的界面, 因此积分面上电流密度  $\mathbf{J} = 0$ , 该项为零, 即:

$$\nabla \cdot \mathbf{A}(\mathbf{x}, t) = \frac{\mu_0}{4\pi} \iiint \frac{1}{r} [\nabla' \cdot \mathbf{J}(\mathbf{x}')] dV'$$

另一方面，对于静电势  $\varphi$ ：

$$\frac{\partial \varphi(\mathbf{x}, t)}{\partial t} = \frac{1}{4\pi\epsilon_0} \iiint \frac{1}{r} \frac{\partial \rho(\mathbf{x}', t')}{\partial t'} dV'$$

故由电荷守恒定律：

$$\nabla' \cdot \mathbf{J}(\mathbf{x}') + \frac{\partial \rho(\mathbf{x}', t')}{\partial t'} = 0$$

得：

$$\nabla \cdot \mathbf{A}(\mathbf{x}, t) + \frac{1}{c^2} \frac{\partial \varphi(\mathbf{x}, t)}{\partial t} = 0$$

即  $\mathbf{A}$  和  $\varphi$  满足 Lorenz 规范。

10. 磁体球的磁矩大小为：

$$m_0 = \frac{4}{3} \pi R_0^3 M_0$$

则旋转磁体球的磁矩为：

$$\mathbf{m} = m_0(\hat{x} \cos \omega t + \hat{y} \sin \omega t)$$

在  $\omega R_0 \ll c$  的近似下产生磁偶极辐射。

由于  $\text{Re}[(\hat{x} + i\hat{y})e^{-i\omega t}] = \hat{x} \cos \omega t + \hat{y} \sin \omega t$ ，故  $\mathbf{m}$  也可以用复数表示为：

$$\mathbf{m} = m_0(\hat{x} + i\hat{y})e^{-i\omega t}$$

又使用直角坐标和球坐标的转换矩阵得到：

$$\mathbf{m} = m_0(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi - i\omega t}$$

则有：

$$\ddot{\mathbf{m}} = (-i\omega)^2 \mathbf{m} = -\omega^2 m_0(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi - i\omega t}$$

故产生的辐射场为：

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0}{4\pi c^2} (\ddot{\mathbf{m}} \times \hat{r}) \times \hat{r} \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 M_0 R_0^3}{3c^2} (\hat{\theta} \cos \theta + i\hat{\phi}) \frac{e^{i\phi - i\omega t + ikr}}{r} \\ \mathbf{E} &= -\frac{\mu_0}{4\pi c} (\ddot{\mathbf{m}} \times \hat{r}) \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 M_0 R_0^3}{3c} (i\hat{\theta} - \hat{\phi} \cos \theta) \frac{e^{i\phi - i\omega t + ikr}}{r} \end{aligned}$$

辐射场的能流密度为：

$$\mathbf{S} = \frac{1}{2\mu_0} \mathbf{E} \times \mathbf{B}^* = \frac{\mu_0 \omega^4 M_0^2 R_0^6}{18c^3} (1 + \cos^2 \theta) \frac{\hat{r}}{r^2}$$

11. 做圆周运动的粒子的电偶极矩为：

$$\mathbf{p} = ea(\hat{x} \cos \omega t + \hat{y} \sin \omega t)$$

辐射场为电偶极辐射。

由于  $\text{Re}[(\hat{x} + i\hat{y})e^{-i\omega t}] = \hat{x} \cos \omega t + \hat{y} \sin \omega t$ ，故  $\mathbf{p}$  也可以用复数表示为：

$$\mathbf{p} = ea(\hat{x} + i\hat{y})e^{-i\omega t}$$

又使用直角坐标和球坐标的转换矩阵得到：

$$\mathbf{p} = ea(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi - i\omega t}$$

则有：

$$\ddot{\mathbf{p}} = (-i\omega)^2 \mathbf{p} = -\omega^2 ea(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi-i\omega t}$$

故产生的辐射场为：

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0 c^2} (\ddot{\mathbf{p}} \times \hat{r}) \times \hat{r} \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 ea}{4\pi} (\hat{\theta} \cos \theta + i\hat{\phi}) \frac{e^{i\phi-i\omega t+ikr}}{r}$$

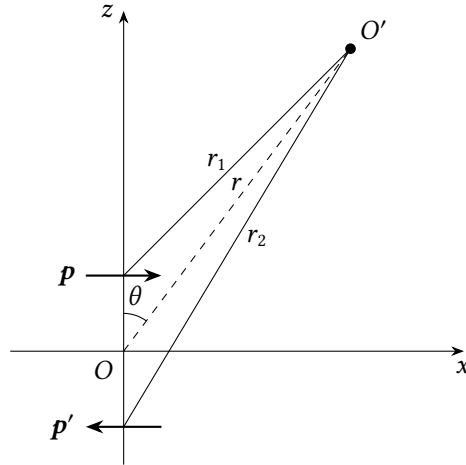
$$\mathbf{B} = \frac{1}{4\pi\epsilon_0 c^3} (\ddot{\mathbf{p}} \times \hat{r}) \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 ea}{4\pi} (\hat{\phi} \cos \theta - i\hat{\theta}) \frac{e^{i\phi-i\omega t+ikr}}{r}$$

辐射场的能流密度为：

$$\mathbf{S} = \frac{1}{2\mu_0} \mathbf{E} \times \mathbf{B}^* = \frac{\mu_0 \omega^4 e^2 a^2}{32\pi^2 c} (1 + \cos^2 \theta) \frac{\hat{r}}{r^2}$$

12. 如图，理想导体平面为  $Oxy$  平面，电偶极子  $\mathbf{p} = pe^{-i\omega t}\hat{x}$  位于  $(0, 0, \frac{a}{2})$ 。

使用电像法，理想导体平面可以等效为电像  $\mathbf{p}' = -pe^{-i\omega t}\hat{x}$  位于  $(0, 0, -\frac{a}{2})$ ，则远处一点  $O'$  的辐射场等于  $\mathbf{p}$  和  $\mathbf{p}'$  的辐射场叠加之和。



在  $r \gg a$  时，如图有：

$$r_1 = \sqrt{r^2 - ra \cos \theta + \left(\frac{a}{2}\right)^2} \approx r - \frac{a}{2} \cos \theta$$

$$r_2 = \sqrt{r^2 + ra \cos \theta + \left(\frac{a}{2}\right)^2} \approx r + \frac{a}{2} \cos \theta$$

且  $\hat{r}_1 = \hat{r}_2 = \hat{r}$ ，以及：

$$e^{\pm ik \frac{a}{2} \cos \theta} = 1 \pm ik \frac{a}{2} \cos \theta$$

球坐标下  $\hat{x} = \hat{r} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ ，则

$$\hat{x} \times \hat{r} = -\hat{\theta} \sin \phi - \hat{\phi} \cos \theta \cos \phi, \quad (\hat{x} \times \hat{r}) \times \hat{r} = -\hat{\theta} \cos \theta \cos \phi + \hat{\phi} \sin \phi$$

故  $\mathbf{p}$  和  $\mathbf{p}'$  产生的磁场为：

$$\mathbf{B} = \frac{\mu_0}{4\pi c} (\ddot{\mathbf{p}} \times \hat{r}) \frac{e^{ik(r-\frac{a}{2}\cos\theta)}}{r} = \frac{\mu_0 \omega^2 p}{4\pi c} (\hat{\theta} \sin \phi + \hat{\phi} \cos \theta \cos \phi) \frac{e^{ikr-i\omega t}}{r} \left(1 - \frac{ika}{2} \cos \theta\right)$$

$$\mathbf{B}' = \frac{\mu_0}{4\pi c} (\ddot{\mathbf{p}}' \times \hat{r}) \frac{e^{ik(r+\frac{a}{2}\cos\theta)}}{r} = -\frac{\mu_0\omega^2 p}{4\pi c} (\hat{\theta} \sin\phi + \hat{\phi} \cos\theta \cos\phi) \frac{e^{ikr-i\omega t}}{r} \left(1 + \frac{ika}{2} \cos\theta\right)$$

总磁场为:

$$\mathbf{B}_0 = \mathbf{B} + \mathbf{B}' = -\frac{\mu_0\omega^2 p}{4\pi c} ika(\hat{\theta} \cos\theta \sin\phi + \hat{\phi} \cos^2\theta \cos\phi) \frac{e^{ikr-i\omega t}}{r}$$

同理,  $\mathbf{p}$  和  $\mathbf{p}'$  产生的电场为:

$$\mathbf{E} = \frac{\mu_0}{4\pi} (\ddot{\mathbf{p}} \times \hat{r}) \times \hat{r} \frac{e^{ik(r-\frac{a}{2}\cos\theta)}}{r} = \frac{\mu_0\omega^2 p}{4\pi} (\hat{\theta} \cos\theta \cos\phi - \hat{\phi} \sin\phi) \frac{e^{ikr-i\omega t}}{r} \left(1 - \frac{ika}{2} \cos\theta\right)$$

$$\mathbf{E}' = \frac{\mu_0}{4\pi} (\ddot{\mathbf{p}}' \times \hat{r}) \times \hat{r} \frac{e^{ik(r+\frac{a}{2}\cos\theta)}}{r} = -\frac{\mu_0\omega^2 p}{4\pi} (\hat{\theta} \cos\theta \cos\phi - \hat{\phi} \sin\phi) \frac{e^{ikr-i\omega t}}{r} \left(1 + \frac{ika}{2} \cos\theta\right)$$

总电场为:

$$\mathbf{E}_0 = \mathbf{E} + \mathbf{E}' = -\frac{\mu_0\omega^2 p}{4\pi} ika(\hat{\theta} \cos^2\theta \cos\phi - \hat{\phi} \cos\theta \sin\phi) \frac{e^{ikr-i\omega t}}{r}$$

考虑到  $k = \frac{\omega}{c}$ , 也可以写为:

$$\mathbf{B}_0 = -\frac{i\mu_0\omega^3 pa}{4\pi c^2} (\hat{\theta} \cos\theta \sin\phi + \hat{\phi} \cos^2\theta \cos\phi) \frac{e^{ikr-i\omega t}}{r}$$

$$\mathbf{E}_0 = -\frac{i\mu_0\omega^3 pa}{4\pi c} ika(\hat{\theta} \cos^2\theta \cos\phi - \hat{\phi} \cos\theta \sin\phi) \frac{e^{ikr-i\omega t}}{r}$$

能流密度为:

$$\mathbf{S}_0 = \frac{1}{2\mu_0} \mathbf{E}_0 \times \mathbf{B}_0 = \frac{\mu_0\omega^6 p^2 a^2}{32\pi^2 c^3} (\cos^2\theta \sin^2\phi + \cos^4\theta \cos^2\phi) \frac{\hat{r}}{r^2}$$

13. 介质球被周期性均匀极化, 其电偶极矩为  $\mathbf{p} = \mathbf{p}_0 e^{-i\omega t}$ 。其中  $\mathbf{p}_0$  为介质球在电场  $\mathbf{E}_0$  极化下的电偶极矩, 计算如下:

由于空间各点都不存在自由电荷分布, 静电势  $\varphi$  在全空间满足 Laplace 方程:

$$\begin{cases} \nabla^2 \varphi_1 = 0, & r < R_0 \\ \nabla^2 \varphi_2 = 0, & r > R_0 \end{cases}$$

边界条件为:

$$\begin{aligned} |\varphi_1|_{r=0} &< +\infty, \quad \varphi_2|_{r \rightarrow +\infty} = -E_0 r \cos\theta \\ r = R_0: \quad \varphi_1 &= \varphi_2, \quad \varepsilon \frac{\partial \varphi_1}{\partial r} = \varepsilon_0 \frac{\partial \varphi_2}{\partial r} \end{aligned}$$

则电势的一般解为:

$$\varphi_1 = \sum_{l=0}^{+\infty} A_l r^l P_l(\cos\theta), \quad \varphi_2 = \sum_{l=0}^{+\infty} (B_l r^l + C_l r^{-l-1}) P_l(\cos\theta)$$

由无穷远处的边界条件:

$$A_0 = B_0 = C_0 = 0, \quad B_1 = -E_0$$

$$B_l = 0, \quad (l \geq 2)$$

考虑到  $r = R_0$  处的边界条件,  $l \geq 2$  时:

$$A_l R_0^l = C_l R_0^{-l-1}$$

$$\varepsilon l A_l R_0^{l-1} = \varepsilon_0 (-l-1) C_l R_0^{-l-2}$$

即需要同时满足:

$$A_l = C_l R_0^{-2l-1} = C_l \frac{\varepsilon_0 (-l-1)}{\varepsilon l} R_0^{-2l-3}$$

只可能是  $A_l = C_l = 0$ 。

而  $l = 1$  时:

$$A_1 R_0 = -E_0 R_0 + C_1 R_0^{-2}$$

$$\varepsilon A_1 = -\varepsilon_0 (E_0 + 2C_1 R_0^{-3})$$

解得:

$$A_1 = -\frac{3\varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0, \quad C_1 = \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0 R_0^3$$

于是得到静电势在全空间的解:

$$\begin{cases} \varphi_1 = -\frac{3\varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0 r \cos \theta, & r < R_0 \\ \varphi_2 = -E_0 r \cos \theta + \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} E_0 R_0^3 \frac{\cos \theta}{r^2}, & r > R_0 \end{cases}$$

介质球内:

$$\mathbf{D}_1 = \varepsilon_0 \mathbf{E}_1 + \mathbf{P} = \varepsilon \mathbf{E}_1$$

则极化强度为:

$$\mathbf{P} = (\varepsilon - \varepsilon_0) \mathbf{E}_1 = -(\varepsilon - \varepsilon_0) \nabla \phi_1 = \frac{3\varepsilon_0(\varepsilon - \varepsilon_0)}{\varepsilon + 2\varepsilon_0} E_0 \hat{\mathbf{z}}$$

电偶极矩为:

$$\mathbf{p}_0 = \frac{4}{3} \pi R_0^3 \mathbf{P}_0 = \frac{4\pi\varepsilon_0(\varepsilon - \varepsilon_0)}{\varepsilon + 2\varepsilon_0} E_0 R_0^3 \hat{\mathbf{z}}$$

因此, 被周期性均匀极化的介质球的电偶极矩为:

$$\mathbf{p} = \mathbf{p}_0 e^{-i\omega t} = \frac{4\pi\varepsilon_0(\varepsilon - \varepsilon_0)}{\varepsilon + 2\varepsilon_0} E_0 R_0^3 e^{-i\omega t} \hat{\mathbf{z}}$$

则有:

$$\ddot{\mathbf{p}} = (-i\omega)^2 \mathbf{p} = -\omega^2 \frac{4\pi\varepsilon_0(\varepsilon - \varepsilon_0)}{\varepsilon + 2\varepsilon_0} E_0 R_0^3 e^{-i\omega t} \hat{\mathbf{z}}$$

由  $\hat{\mathbf{z}} = \hat{\mathbf{r}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$  得:

$$\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \sin \theta, \quad (\hat{\mathbf{z}} \times \hat{\mathbf{r}}) \times \hat{\mathbf{r}} = \hat{\boldsymbol{\theta}} \sin \theta$$

故产生的辐射场为:

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0 c^2} (\ddot{\mathbf{p}} \times \hat{\mathbf{r}}) \times \hat{\mathbf{r}} \frac{e^{ikr}}{r} = -\frac{\omega^2 E_0 R_0^3}{c^2} \left( \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} \right) \hat{\boldsymbol{\theta}} \sin \theta \frac{e^{ikr-i\omega t}}{r}$$

$$\mathbf{B} = \frac{1}{4\pi\varepsilon_0 c^3} (\ddot{\mathbf{p}} \times \hat{\mathbf{r}}) \frac{e^{ikr}}{r} = -\frac{\omega^2 E_0 R_0^3}{c^3} \left( \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} \right) \hat{\boldsymbol{\phi}} \sin \theta \frac{e^{ikr-i\omega t}}{r}$$

辐射场的能流密度为:

$$\mathbf{S} = \frac{1}{2\mu_0} \mathbf{E} \times \mathbf{B}^* = \frac{\varepsilon_0 \omega^4 E_0^2 R_0^6}{2c^3} \left( \frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0} \right)^2 \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}$$

## 补充题

1. 以线圈所在平面法向为  $\hat{x}$  方向，旋转轴为  $\hat{z}$  方向，线圈的磁矩大小为：

$$m_0 = \pi a^2 I$$

则旋转线圈的磁矩为：

$$\mathbf{m} = m_0(\hat{x} \cos \omega t + \hat{y} \sin \omega t)$$

在  $\omega a \ll c$  的近似下产生磁偶极辐射。

由于  $\text{Re}[(\hat{x} + i\hat{y})e^{-i\omega t}] = \hat{x} \cos \omega t + \hat{y} \sin \omega t$ ，故  $\mathbf{m}$  也可以用复数表示为：

$$\mathbf{m} = m_0(\hat{x} + i\hat{y})e^{-i\omega t}$$

又使用直角坐标和球坐标的转换矩阵得到：

$$\mathbf{m} = m_0(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi - i\omega t}$$

则有：

$$\ddot{\mathbf{m}} = (-i\omega)^2 \mathbf{m} = -\omega^2 m_0(\hat{r} \sin \theta + \theta \cos \theta + i\hat{\phi})e^{i\phi - i\omega t}$$

故产生的辐射场为：

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0}{4\pi c^2} (\ddot{\mathbf{m}} \times \hat{r}) \times \hat{r} \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 a^2 I}{4c^2} (\hat{\theta} \cos \theta + i\hat{\phi}) \frac{e^{i\phi - i\omega t + ikr}}{r} \\ \mathbf{E} &= -\frac{\mu_0}{4\pi c} (\ddot{\mathbf{m}} \times \hat{r}) \frac{e^{ikr}}{r} = \frac{\mu_0 \omega^2 a^2 I}{4c} (i\hat{\theta} - \hat{\phi} \cos \theta) \frac{e^{i\phi - i\omega t + ikr}}{r} \end{aligned}$$

2. 以线圈所在平面法向为  $\hat{z}$  方向，圆形线圈的磁矩为：

$$\mathbf{m} = \pi a^2 I e^{-i\omega t} \hat{z}$$

在  $ka \ll 1$ ，即  $\omega a \ll c$  的近似下产生磁偶极辐射。

则有：

$$\ddot{\mathbf{m}} = (-i\omega)^2 \mathbf{m} = -\omega^2 \pi a^2 I e^{-i\omega t} \hat{z}$$

故产生的辐射场为：

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0}{4\pi c^2} (\ddot{\mathbf{m}} \times \hat{r}) \times \hat{r} \frac{e^{ikr}}{r} = -\frac{\mu_0 \omega^2 a^2 I}{4c^2} \hat{\phi} \sin \theta \frac{e^{ikr - i\omega t}}{r} \\ \mathbf{E} &= -\frac{\mu_0}{4\pi c} (\ddot{\mathbf{m}} \times \hat{r}) \frac{e^{ikr}}{r} = -\frac{\mu_0 \omega^2 a^2 I}{4c} \hat{\theta} \sin \theta \frac{e^{ikr - i\omega t}}{r} \end{aligned}$$

辐射场的能流密度为：

$$\mathbf{S} = \frac{1}{2\mu_0} \mathbf{E} \times \mathbf{B}^* = \frac{\mu_0 \omega^4 a^4 I^2}{32c^3} \frac{\sin^2 \theta}{r^2} \hat{r}$$

辐射功率的角分布为：

$$\frac{dP}{d\Omega} = \lim_{r \rightarrow +\infty} r^2 (\mathbf{S} \cdot \hat{r}) = \frac{\mu_0 \omega^4 a^4 I^2}{32c^3} \sin^2 \theta$$

总辐射功率为：

$$P = \oint \frac{dP}{d\Omega} d\Omega = \int_0^{2\pi} d\phi \int_0^\pi \frac{\mu_0 \omega^4 a^4 I^2}{32c^3} \sin^2 \theta \cdot \sin \theta d\theta = \frac{\pi \mu_0 \omega^4 a^4 I^2}{12c^3}$$